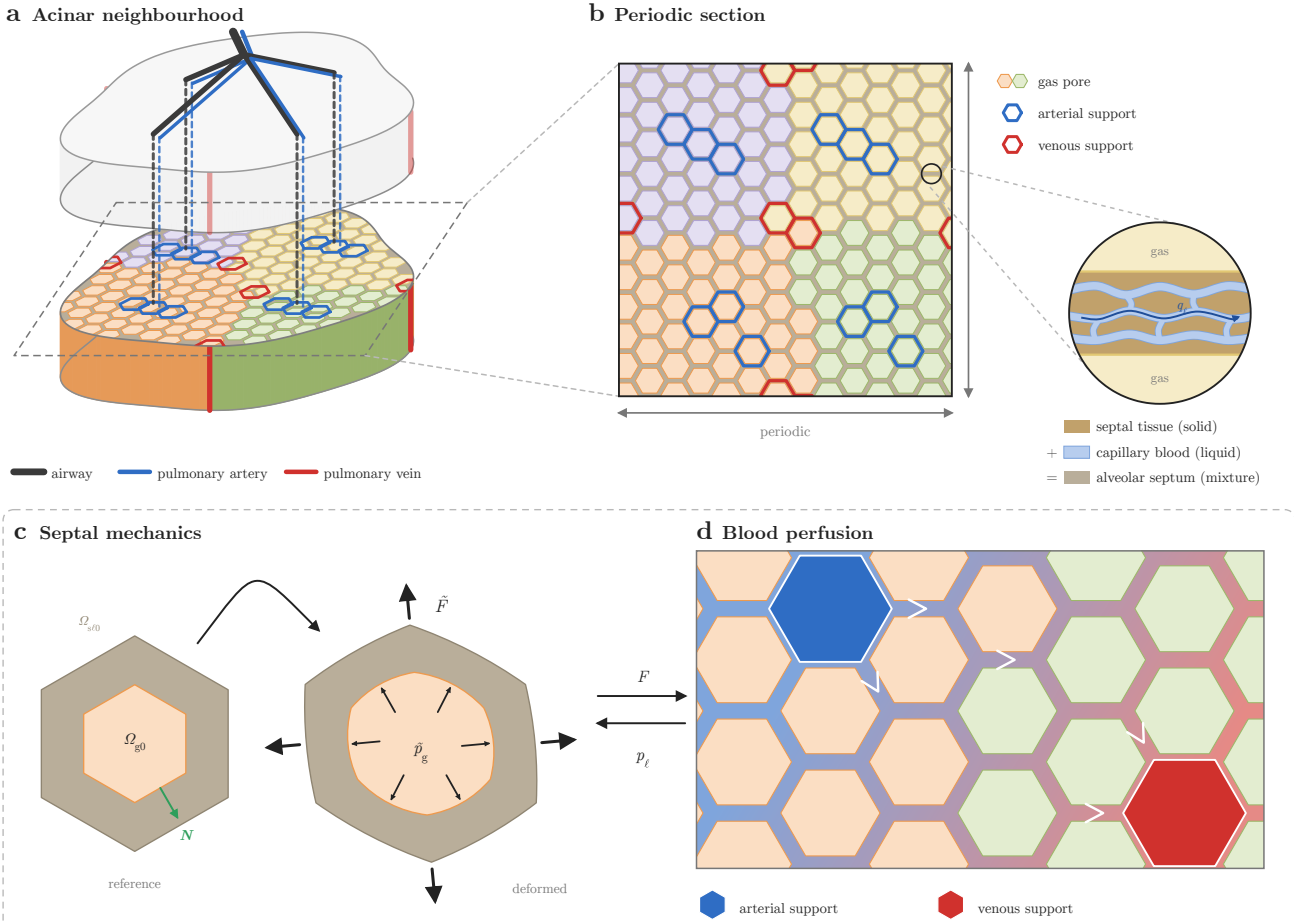

a Acinar neighbourhood
airway
pulmonary artery
pulmonary vein
b Periodic section
gas pore
arterial support
venous support
gas
q_ℓ
gas
periodic
septal tissue (solid)
+ capillary blood (liquid)
= alveolar septum (mixture)
c Septal mechanics
$\Omega_{s/0}$
Ω_{g0}
N
reference
$\tilde{F}$
$\tilde{p}_g$
deformed
F
p_ℓ
d Blood perfusion
arterial support
venous support